Agentic AI Flight Delay Risk Analyser

Presented by:

M.Janani priya-24071A6736

csd-A

PROBLEM STATEMENT

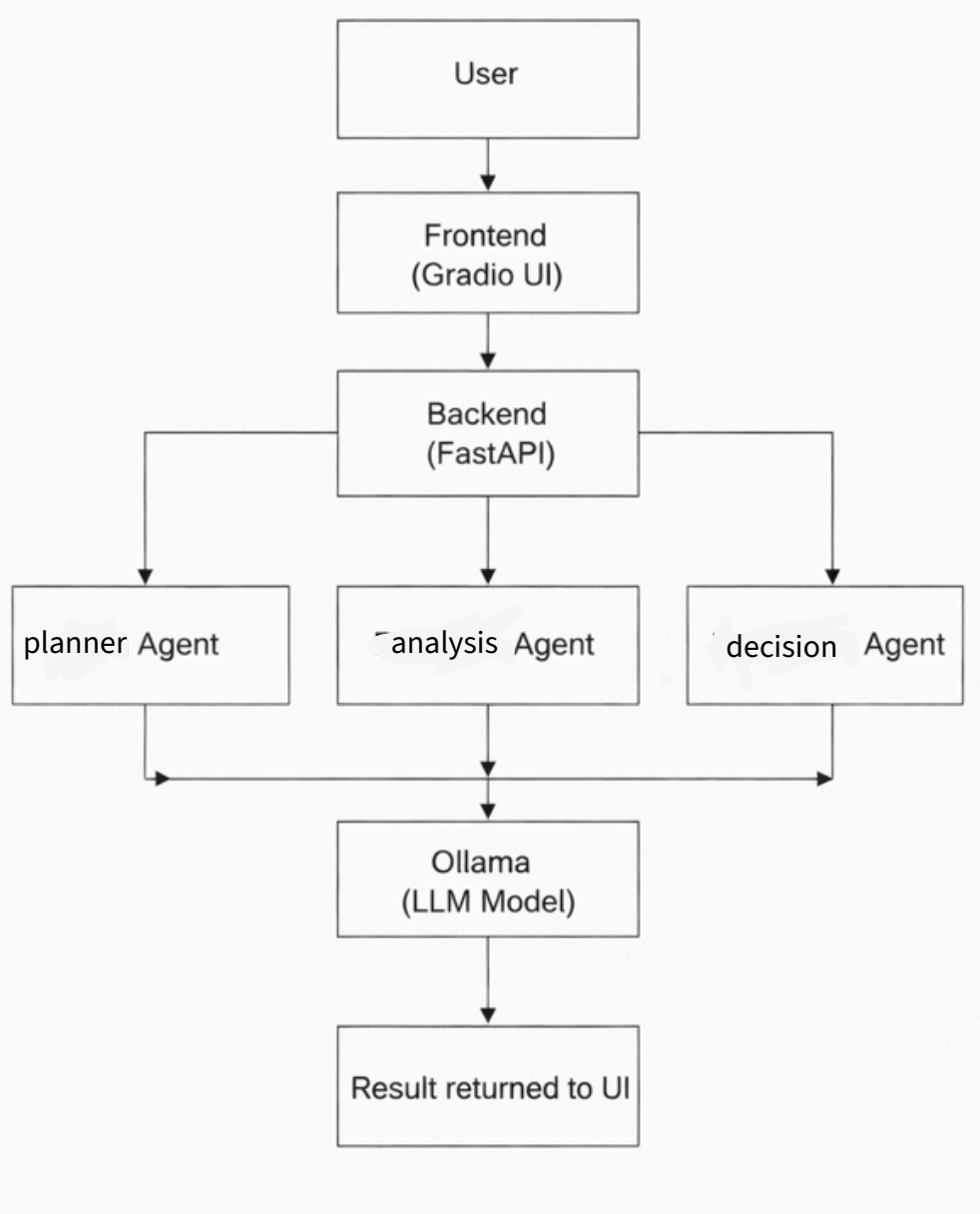
- Flight delays are common in aviation operations.
- Delays are caused by factors such as weather conditions, airport congestion, and aircraft arrival delays.
- Airport operators must quickly analyze multiple operational conditions.
- Current systems rely on manual monitoring and separate tools.
- There is a need for an AI-based system to analyze delay risks and support faster decision-making.

PROPOSED SOLUTION

- Develop an Agentic AI system to analyze flight delay risks.
- Use multiple AI agents to collaborate and analyze operational factors.
- The Planner Agent decides which conditions should be analyzed.
- The Analysis Agent collects factors such as weather, airport traffic, and aircraft delay.
- The Decision Agent uses a local LLM to evaluate conditions and predict delay risk
- The system provides delay risk prediction and recommended operational actions.

.

ARCHITECTURE DIAGRAM



- User inputs flight details through Streamlit UI
- Backend processes request using LangGraph workflow
- Agents collaborate to analyze delay factors
- LLM generates delay prediction and recommendation

FUTURE SCOPE

- Integrate real-time aviation APIs to fetch live flight and weather data.
- Use historical flight datasets to improve delay prediction accuracy.
- Implement machine learning models for advanced delay forecasting.
- Add automated airport resource management for gates and runways.
- Deploy the system as a cloud-based application for real-world airport operations.

SCREENSHOTS

AI Recommendation

Based on the analysis, I predict a moderate to high risk of Flight BQ203 experiencing delays due to both weather conditions (heavy rain) causing potential ground operations disruptions and an unusually high traffic volume leading to airport congestion. Historical data suggest that these combined factors can result in flight departure delays during the winter season at JFK International Airport.

Recommended actions include:

- Prioritizing de-icing procedures for Flight BQ203 and any other aircraft scheduled shortly after to mitigate weather impacts on ground operations, ensuring safety remains paramount without unnecessarily delaying departures (if conditions allow).
- Communicating with the airline's gate staff about potential delays due to traffic congestion. This will facilitate a more informed decision making process regarding departure timing and possible adjustments in boarding processes or flight scheduling when necessary, based on real-time data assessment from JFK Airport operations center.
- Providing regular updates via the airport's communication channels to inform passengers of any changes as early warning systems are crucial for maintaining trust and efficiency during disruptions in flights like BQ203 departure delays, helping manage expectations proactively rather than reactively following incidents.
- Consultation with the airport's meteorology department should be advised to get a better understanding of weather trends that might further impact flight operations today and advising on possible contingency plans such as gate transfers or holding patterns if delays become significant, always maintaining safety protocols in accordance.
- Coordinating with the air traffic control (ATC) unit to manage runway allocation effectively during peak congestion periods may help reduce delay risks for BQ203 and subsequent flights through dynamic scheduling adjustments as needed while ensuring compliance with FAA regulations regarding aircraft separation minimums.
- If delays are unavoidable due to the extreme nature of weather or traffic, I would advise airlines to consider strategic rerouting options that could mitigate delay times and preserve fuel efficiency by selecting alternate flight paths less affected by localized conditions when feasible within safety margins.

SUGGESTED IMPROVEMENTS

- Integrate real-time weather and flight data APIs for more accurate analysis.
- Improve the agent collaboration workflow for faster decision making.
- Optimize the LLM response time to reduce system latency.
- Enhance the user interface with better visualization of delay factors.
- Add predictive analytics using historical flight data to improve delay forecasting.

Conclusion

- Developed an intelligent multi-agent system for flight delay risk prediction
- Successfully integrated Planner, Analysis, and Decision agents using LangGraph
- Leveraged a local LLM (Ollama – Phi-3) for real-time decision-making
- Reduced dependency on manual monitoring and fragmented tools
- Demonstrated how AI can improve operational efficiency in aviation
- Provided actionable recommendations, not just predictions
- Designed a scalable and modular architecture for future expansion
- Ensured privacy and low latency by running entirely on local infrastructure
- Showcased practical application of Agentic AI in real-world scenarios
- Established a foundation for advanced predictive analytics in aviation systems

THANK YOU